

Software Requirements Specification Document

System Overview

- An **online** DVD lending system for students and faculty
- Requires users to have own accounts / IDs for DVD tracking and security
- Keeps a checkout history to have a detailed log of which DVDs are checked out (potentially charge users who damage or lose DVDs)
- Tracks which DVDs are available and unavailable (being borrowed by students / faculty)

Stakeholders and User Roles

- Borrowers (students and faculty): borrow and return DVDs.
- Staff Members: Manage and handle DVD inventory and borrowing rules.
- System administrator: Oversees the system and reports.

Functional Requirements as User Stories

- Frontend
 - As a Borrower, I want to be able to browse through a catalog of DVDs that I can sort by title, genre, or availability so that I can easily find a DVD I want to borrow.
 - As a Borrower, I want to be able to click on a DVD and see the details of it, including summary, as well as have options to check it out so that I can easily borrow it online.
 - As a Borrower, I want to be able to view my checked out DVDs and see the due date, so that I always know when it has to be returned by.
- Backend
 - As a Staff Member, I want to be able to process every request from the frontend and return a response within 5 seconds so that all information is up-to-date and users are not dissatisfied with response time.
 - As a System Administrator, I want to be the only group able to access the Backend so that access to internal data can be controlled.
- Database
 - As a Staff Member, I want the data to be consistent across both the Frontend and the Backend so that unavailable DVDs are not borrowed a second time, or available DVDs are not marked as unavailable.
- Authentication & Authorization
 - As a Borrower, I should be able to use my ID to borrow DVD and have some authentication like facial recognition or otp.
- Monitoring, Logging & Analytics
 - The system should record all the DVD status and location. Also record all the past borrower history.

Non-Functional Requirements

- Scanning and accessing the user ID and DVDs shouldn't take more than 5 seconds
- Users accounts should be secure with no data leaks as there will be potential purchases with credit cards and important private information
- User passwords encrypted before storage
- Code must have 80% unit test coverage

Explicit Assumptions and Exclusions

- DVDs that go missing while borrowed will be marked as "lost"
 - Administrators can remove the entry from the database
 - Administrators can order replacement DVDs and add to database
- All data on which DVDs are borrowed or available will be up-to-date